GMI German-Malaysian Institute

DIPLOMA PROGRAMME

PRACTICAL TEST 1

Academic Period : JULY 2026

Subject	**:**	**CRYPTOGRAPHY**
Code	**:**	**CBS 2372**
Duration	**:**	**3 HOURS**
Total Marks	**:**	**50 MARKS**

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Do not open this question booklet until you are told to do so by the invigilator.
3. Read the specific instructions and questions carefully before answering.

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Student's I/D No.	**:**	CBS24070208
Semester/Trade/Group	**:**	SEM 5 / DCBS / CBS 1
Date	**:**	21 / 10 / 2026

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	Affine cipher - To calculate modular multiplicative inverse and decrypt by using Excel (include the calculation)	2	4	P3	15			
2.	Hill cipher (2 x 2, row vector) - To encrypt and decrypt by using Excel (include the calculation)	2	4	P3	15			
3.	Playfair cipher - To encrypt by using Excel (include the calculation)	2	4	P3	10			
4.	Columnar Transposition technique - To encrypt and decrypt by using Excel (include the calculation)	2	4	P3	10			
				Total	50			
				Total Marks	/50			
	Total		**PO4**		/50			
				Signature	Jumamelissa Sani			

Learning Objectives:

1. To calculate modular multiplicative inverse
2. To encrypt and decrypt:
    a. Affine cipher technique (decrypt)
    b. Hill cipher technique (2 x 2, row vector)
    c. Playfair cipher technique (encrypt)
    d. Columnar Transposition technique (encrypt and decrypt)

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To do this Practical Test, you should open the Microsoft Excel and start to follow the question given. These tasks are based on the various Classical Cipher algorithm techniques.

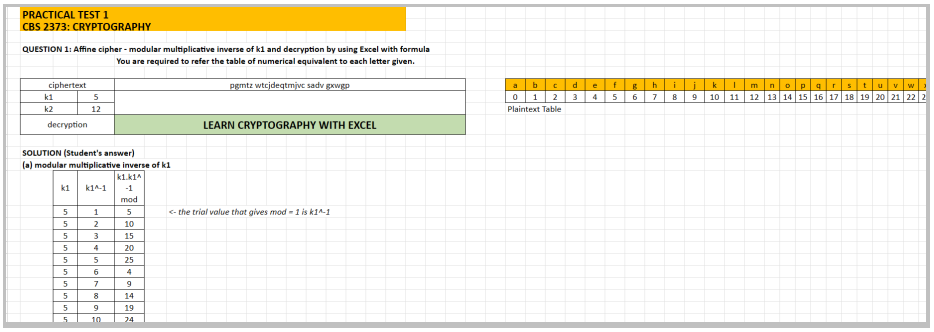
You are required to submit the answer in word document to PDF FORMAT and EXCEL FILE with mod calculation.

All answers are based on the Excel solution given.

Tasks	Questions	Marks
Affine cipher: To calculate modular multiplicative inverse and decrypt by using Excel (include the calculation)	**QUESTION 1:** Calculate the modular multiplicative inverse of k1 and give the decryption answer by using **Excel including the calculation**. **(inverse 5 marks + decryption 10 marks)**	**15**

Ciphertext	pgmtz wtcjdeqtmjvc sadv gxwgp
k1	5
k2	12
k1^-1	21
Decryption	Learn Cryptography With Excel

Screen shot the Excel result:



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QUESTION 1: Affine cipher - modular multiplicative inverse of k1 and decryption by using Excel with formula
You are required to refer the table of numerical equivalent to each letter given.

ciphertext	pgmtz wtcjdeqtmjvc sadv gxwgp
k1	5
k2	12
decryption	LEARN CRYPTOGRAPHY WITH EXCEL

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22

Plaintext Table

SOLUTION (Student's answer)
(a) modular multiplicative inverse of k1

k1	k1^-1	k1.k1^-1 mod
5	1	5
5	2	10
5	3	15
5	4	20
5	5	25
5	6	4
5	7	9
5	8	14
5	9	19
5	10	24
5	20	22
5	21	1
5	22	6
5	23	11
5	24	16
5	25	21
5	26	0

<- the trial value that gives mod = 1 is k1^-1

pgmtz wtcjdeqtmjvc sadv gxwgp

(b) decryption

ciphertext	P	G	M	T	Z		W	T	C	J	D	E	Q	T	M	J	V	C		S	A	D	V		G	X	W	G	P
	15	6	12	19	25		22	19	2	9	3	4	16	19	12	9	21	2		18	0	3	21		6	23	22	6	15

k1	5
k2	12
k1^-1	21

<- k1^-1 from the mod table in (a)

decryption

(x-k2)	3	-6	0	7	13		10	7	-10	-3	-9	-8	4	7	0	-3	9	-10		6	-12	-9	9		-6	11	10	-6	3
k1^-1(x-k2)	63	-126	0	147	273		210	147	-210	-63	-189	-168	84	147	0	-63	189	-210		126	-252	-189	189		-126	231	210	-126	63
k1^-1(x-k2) mod 26	11	4	0	17	13		2	17	24	15	19	14	6	17	0	15	7	24		22	8	19	7		4	23	2	4	11
plaintext	L	E	A	R	N		C	R	Y	P	T	O	G	R	A	P	H	Y		W	I	T	H		E	X	C	E	L

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Plaintext Table

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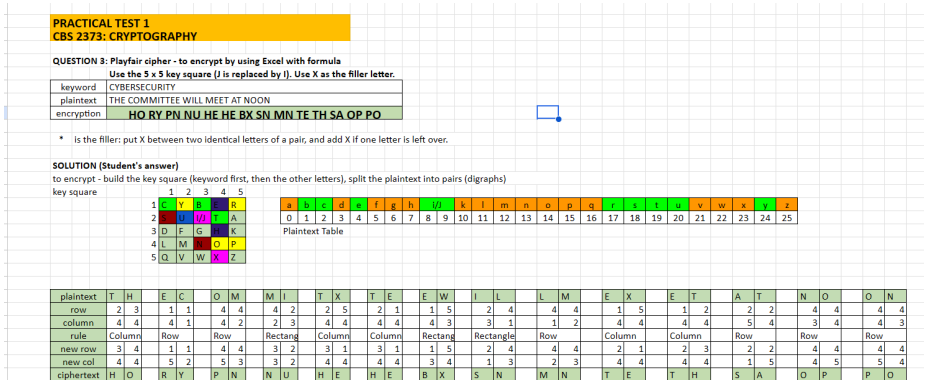
Hill cipher (2 x 2, row vector): To encrypt and decrypt by using Excel (include the calculation)	**QUESTION 2:** Give the encryption answer by using **Excel including the calculation**. Use the row vector method: ciphertext = plaintext x key mod 26. **(8 marks)** Give the decryption answer by using **Excel including the calculation**. Use the row vector method: plaintext = ciphertext x key^-1 mod 26. **(7 marks)** **Screen shot the Excel result:**	**15**

Plaintext	we study classical ciphers today		
key	3	2	
	5	7	
Encryption	?		

Ciphertext	?		
Key^-1	?	?	
	?	?	
Decryption	we study classical ciphers today		

Playfair cipher: To encrypt by using Excel (include the calculation)	**QUESTION 3:** Give the encryption answer by using **Excel including the calculation**. Use the 5 x 5 key square (J is replaced by I). Use X as the filler letter. **(10 marks)** **Screen shot the Excel result:**	**10**

keyword	CYBERSECURITY
Plaintext	THE COMMITTEE WILL MEET AT NOON
Encryption	HO RY PN NU HE HE BX SN MN TE TH SA OP PO



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QUESTION 3: Playfair cipher - to encrypt by using Excel with formula
Use the 5 x 5 key square (J is replaced by I). Use X as the filler letter.

keyword	CYBERSECURITY
plaintext	THE COMMITTEE WILL MEET AT NOON
encryption	HO RY PN NU HE HE BX SN MN TE TH SA OP PO

* Is the filler: put X between two identical letters of a pair, and add X if one letter is left over.

SOLUTION (Student's answer)
to encrypt - build the key square (keyword first, then the other letters), split the plaintext into pairs (digraphs)
key square

	1	2	3	4	5
1	C	Y	B	E	R
2	I/J	I/J	T	A	
3	D	F	G	H	K
4	L	M	N	O	P
5	Q	V	W	X	Z

a	b	c	d	e	f	g	h	i/j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Plaintext Table

plaintext	T	H		E	C		O	M		M	I		T	X		T	E		E	W		I	L		L	M		E	X		E	T		A	T		N	O		O	N	
row	2	3		1	1		4	4		4	2		2	5		2	1		1	5		2	4		4	4		1	5		1	2		2	2		4	4		4	4	
column	4	4		4	1		4	2		2	3		4	4		4	4		4	3		3	1		1	2		4	4		4	4		5	4		3	4		4	3	
rule	Column			Row			Row			Rectang			Column			Column			Rectang			Rectangle			Row			Column			Column			Row			Row			Row		
new row	3	4		1	1		4	4		3	2		3	1		3	1		1	5		2	4		4	4		2	1		2	3		2	2		4	4		4	4	
new col	4	4		5	2		5	3		3	2		4	4		4	4		3	4		1	3		2	3		4	4		4	4		1	5		4	5		5	4	
ciphertext	H	O		R	Y		P	N		N	U		H	E		H	E		B	X		S	N		M	N		T	E		T	H		S	A		O	P		P	O	

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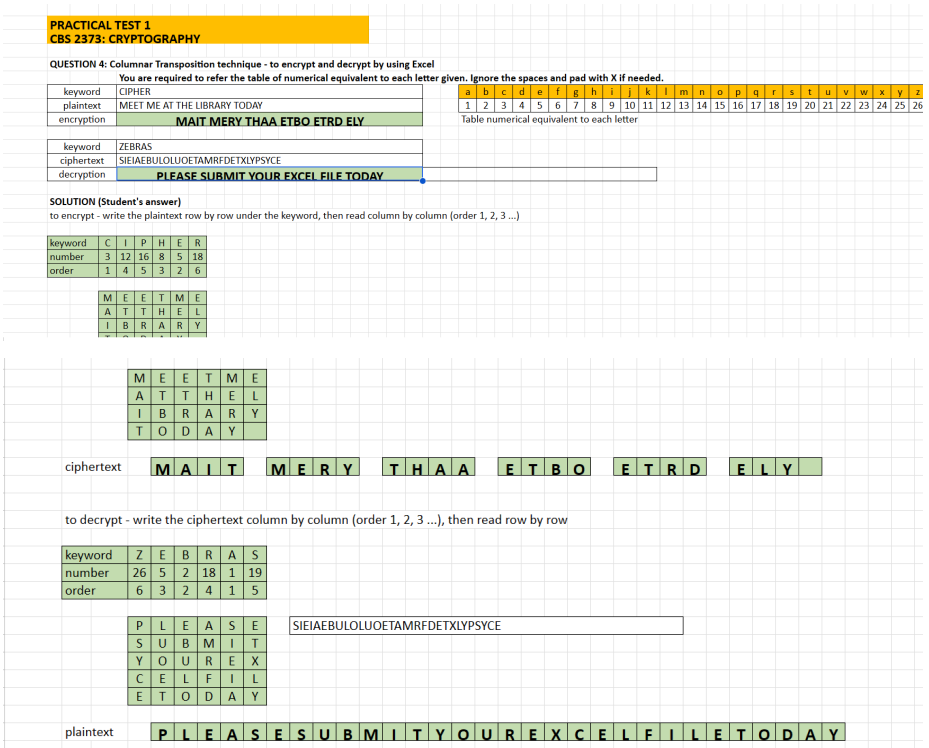
Columnar Transposition technique: To encrypt and decrypt by using Excel (include the calculation)	**QUESTION 4:** Give the encryption answer by using **Excel including the calculation**. Ignore the spaces and pad the last row with X if needed. **(5 marks)**	**10**

keyword	CIPHER
Plaintext	MEET ME AT THE LIBRARY TODAY
Encryption	MAIT MERY THAA ETBO ETRD ELY

Give the decryption answer by using **Excel including the calculation**.
Write the ciphertext column by column in the order of the keyword.
(5 marks)

keyword	ZEBRAS
Ciphertext	SIEIAEBULOLUOETAMRFDETXLYPSYCE
Decryption	Please Submit Your Excel File Today

Screen shot the Excel result:



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QUESTION 4: Columnar Transposition technique - to encrypt and decrypt by using Excel
You are required to refer the table of numerical equivalent to each letter given. Ignore the spaces and pad with X if needed.

keyword	CIPHER
plaintext	MEET ME AT THE LIBRARY TODAY
encryption	MAIT MERY THAA ETBO ETRD ELY

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Table numerical equivalent to each letter

keyword	ZEBRAS
ciphertext	SIEIAEBULOLUOETAMRFDETXLYPSYCE
decryption	PLEASE SUBMIT YOUR EXCEL FILE TODAY

SOLUTION (Student's answer)
to encrypt - write the plaintext row by row under the keyword, then read column by column (order 1, 2, 3 ...)

keyword	C	I	P	H	E	R
number	3	12	16	8	5	18
order	1	4	5	3	2	6

M	E	E	T	M	E
A	T	T	H	E	L
I	B	R	A	R	Y
T	O	D	A	Y	

ciphertext: M A I T | M E R Y | T H A A | E T B O | E T R D | E L Y

to decrypt - write the ciphertext column by column (order 1, 2, 3 ...), then read row by row

keyword	Z	E	B	R	A	S
number	26	5	2	18	1	19
order	6	3	2	4	1	5

P	L	E	A	S	E
S	U	B	M	I	T
Y	O	U	R	E	X
C	E	L	F	I	L
E	T	O	D	A	Y

SIEIAEBULOLUOETAMRFDETXLYPSYCE

plaintext: P L E A S E S U B M I T Y O U R E X C E L F I L E T O D A Y

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